Hi JF, please prepare the workbook below using our existing analysis. We are moving to the output phase now so we need a consistent set of tables to check the MCA comparison and document the six AI-Enabled Expansion occupations. I will use these outputs to review and interpret the results.

What to submit

Submit AI_Research_Technical_Outputs.xlsx with exactly five tabs in this order: Occupation_Master, Archetype_Comparison, Sector_Comparison, Six_Occupations, and Checks. Also submit the notebook or script that generates the tables, with the input filenames and run instructions recorded at the top.

We want to have the output as excel so others can easily access it.

Working rules

- Use the current analysis and existing archetype classifications. Apply the same thresholds to both samples; do not calculate separate MCA thresholds. If the current classifications use different thresholds, flag this before generating a revised comparison.
- Keep SOC codes as text. Count MCA careers using a stable career identifier; if none exists, record the field used to identify distinct entries.
- Separate counts of MCA career entries from counts of unique SOC occupations. Do not duplicate an MCA entry because a join creates multiple rows.
- List unresolved multiple mappings in Checks. Do not pick a SOC code arbitrarily or change an existing mapping rule. If a calculation depends on an unresolved choice, mark it pending and complete the unaffected outputs.
- Use existing sector assignments. If the two datasets use incompatible sector groupings, flag the issue instead of inventing a new crosswalk.
- Keep genuine zero scores as zero. For unavailable information enter Not available; for a zero denominator enter N/A. Do not treat missing data as zero.
- Show counts as integers and percentage values to one decimal place. Calculate percentage values on a 0–100 scale, not as Excel percentage formatting applied to values already multiplied by 100.
- No new datasets or analysis runs are required. No findings summary is required. Record questions and discrepancies in Checks.

1 Occupation Master

Tab name: Occupation_Master. One row per unique SOC occupation in the existing O*NET/SOC comparison sample, including occupations with no MCA match. Sort by SOC_code ascending.

Above the table, record the input filenames, dataset versions or dates, exact variable names for both axes, archetype thresholds and boundary rules, selected current run, sector source, and existing mapping rule. Identify whether each axis uses raw values, ranks, or transformed values.

Column name	Value or calculation
SOC_code	Occupation code used in the existing analysis.
SOC_title	Corresponding occupation title.
Sector	Sector assigned in the existing analysis.
AI_potential_score	Existing value used for the potential axis.
AI_observed_score	Existing value used for the observed-use axis.
AI_archetype	Existing classification in the current run.
MCA_entry_count	Number of distinct MCA career entries mapped to this SOC code under the existing mapping rule.
In_MCA	Yes if MCA_entry_count > 0; otherwise No.
Potential_imputed	Yes, No, or Unknown.
Observed_imputed	Yes, No, or Unknown.

Keep the comparison sample fixed. List any MCA-matched SOC codes outside this sample in Checks. Do not silently add them to the comparison denominator or drop them from the MCA result.

Use the original MCA entry-level data for MCA-entry calculations in the next two tabs. Occupation_Master alone is not sufficient to reproduce those counts if mappings are unresolved or an MCA entry has multiple SOC codes.

2 Archetype Comparison

Tab name: Archetype_Comparison. One row per existing archetype, followed by Total. Include every existing archetype, even if its count is zero in one sample. Use the current archetype order.

Column name	Value or calculation
AI_archetype	Existing archetype name.
SOC_count	Number of comparison-sample SOC occupations in the archetype.
SOC_share_pct	SOC_count ÷ total classified comparison-sample SOC occupations × 100.
MCA_entry_count	Number of distinct classified MCA entries in the archetype.
MCA_entry_share_pct	MCA_entry_count ÷ total classified MCA entries × 100.
MCA_unique_SOC_count	Number of distinct classified MCA-matched SOC codes in the archetype.
MCA_unique_SOC_share_pct	MCA_unique_SOC_count ÷ total distinct classified MCA-matched SOC codes × 100.
Entry_gap_pp	MCA_entry_share_pct − SOC_share_pct.
Unique_SOC_gap_pp	MCA_unique_SOC_share_pct − SOC_share_pct.

Above the table, display the three actual denominators: comparison SOC occupations, MCA entries, and unique MCA-matched SOC occupations. Identify any exclusions and link them to Checks.

The Total row should show total counts, 100.0 for each valid share column, and 0.0 for each gap column. A gap is measured in percentage points, abbreviated pp.

Reconciliation targets: 537 comparison occupations with 139 Active Transformation; 852 MCA entries with 322 Active Transformation. These are prior results to check, not values to force the output to match.

3 Sector Comparison

Tab name: Sector_Comparison. One row per sector, followed by Total. Use the same sector grouping for both samples. Sort by MCA_entries_active_count from largest to smallest; break ties alphabetically.

Column name	Value or calculation
Sector	Existing sector name. Use Unknown for unassigned sectors and flag them.
SOC_total	All classified comparison-sample occupations in this sector.
SOC_active_count	Those classified as Active Transformation.
SOC_active_pct	SOC_active_count ÷ SOC_total × 100.
MCA_entries_total	All distinct classified MCA entries in this sector.
MCA_entries_active_count	Those classified as Active Transformation.
MCA_entries_active_pct	MCA_entries_active_count ÷ MCA_entries_total × 100.
MCA_unique_SOC_total	Distinct classified MCA-matched SOC codes in this sector.
MCA_unique_SOC_active_count	Those classified as Active Transformation.
MCA_unique_SOC_active_pct	MCA_unique_SOC_active_count ÷ MCA_unique_SOC_total × 100.
Within_sector_gap_pp	MCA_unique_SOC_active_pct − SOC_active_pct.
Share_of_MCA_active_entries_pct	Sector MCA_entries_active_count ÷ all MCA Active Transformation entries × 100.

Calculate total-row percentages from summed counts. Do not average sector percentages. The last column sums to 100.0, allowing rounding. The three within-sector percentage columns do not sum to 100 across sectors.

Each counted record must belong to one sector for additive totals. Flag multi-sector assignments instead of double counting. If consistent sector assignment is unresolved, mark the affected comparison pending.

4 Six Occupations

Tab name: Six_Occupations. Create exactly six rows, in this order, using the exact corresponding SOC codes:

- Fine Artists, Including Painters, Sculptors, and Illustrators
- Retail Salespersons
- Bookkeeping, Accounting, and Auditing Clerks
- Medical Transcriptionists
- Administrative Law Judges, Adjudicators, and Hearing Officers
- Judges, Magistrate Judges, and Magistrates

Column name	Value or calculation
SOC_code and SOC_title	Two separate columns with the exact code and title.
AI_potential_score and AI_observed_score	Two separate columns with values from the current run.
Current_archetype	Classification in the current run.
MCA_entry_count	Number of distinct mapped MCA entries.
MCA_titles	Mapped career titles separated by semicolons. If none, enter None.
Archetype_[run_name]	One separate column per existing comparable run; enter that run's classification, or Not available.
Expansion_runs	Number of included runs classifying the occupation as AI-Enabled Expansion.
Available_runs	Number of included runs with a classification for this occupation.
Expansion_frequency_pct	Expansion_runs ÷ Available_runs × 100; N/A if Available_runs = 0.
Data_flag	Missing score, imputation, uncertain mapping, or None identified. List all applicable flags.

Below the table, add a run key with columns Run_name, Input_version, Potential_variable, Observed_variable, Threshold_rule, and Change_from_current_run. Use only existing runs that use the same archetype meanings. List excluded runs and the reason in Checks. If only one run exists, report one; do not describe the result as stable across specifications.

5 Checks and submission

Tab name: Checks. Create a table with five columns in this order: Check, Expected, Actual, Pass/Flag, Notes. Use Pass only when the check is satisfied; otherwise use Flag and explain the discrepancy or unavailable information.

Check to perform	Expected result
Comparison sample count	Expected: 537 total occupations.
Comparison Active Transformation count	Expected: 139 occupations.
MCA classified entry count	Expected: 852 entries.
MCA Active Transformation count	Expected: 322 entries.
Archetype count totals	Expected: each count column sums to its declared denominator.
Archetype share totals	Expected: each share column sums to 100%, within rounding tolerance of 0.1 × number of archetypes percentage points.
Sector count totals	Expected: each total agrees with its corresponding archetype-table count.
Unique master codes	Expected: zero duplicate SOC codes in Occupation_Master.
Common thresholds	Expected: identical definitions, thresholds, transformations, and boundary rules across the two samples.
Six occupation rows	Expected: exactly six requested occupations, with no duplicates.
Expansion run counts	Expected: Expansion_runs ≤ Available_runs; all counted runs appear in the run key.

Below the checks table, add an exception register with columns MCA_entry_ID, MCA_title, SOC_code_or_candidates, Issue, Treatment_in_outputs. Include entries with no SOC match, multiple matches, missing archetype, missing or conflicting sector, and matched SOC codes outside the comparison sample. For a run-level issue, use N/A in entry fields and name the run in Issue.

File Formatting

- Please freeze each table's header row, enable filters on data rows, and keep total rows outside filter ranges. Wrap long titles and widen columns so values are readable.